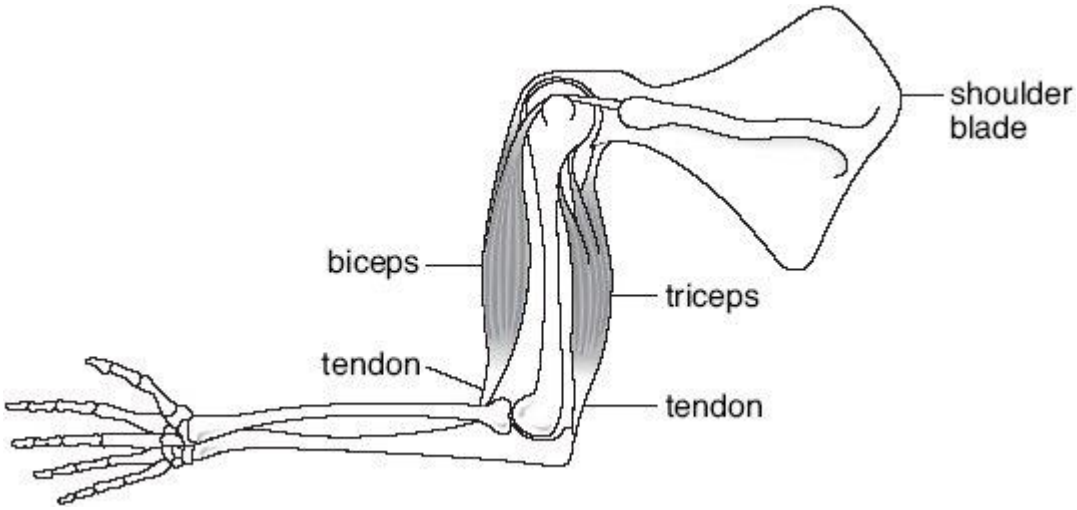
Q1. The diagram below shows muscles and bones of a human arm.



(a) Why is it important that the tendons do **not** stretch?

.....

.....

1 mark

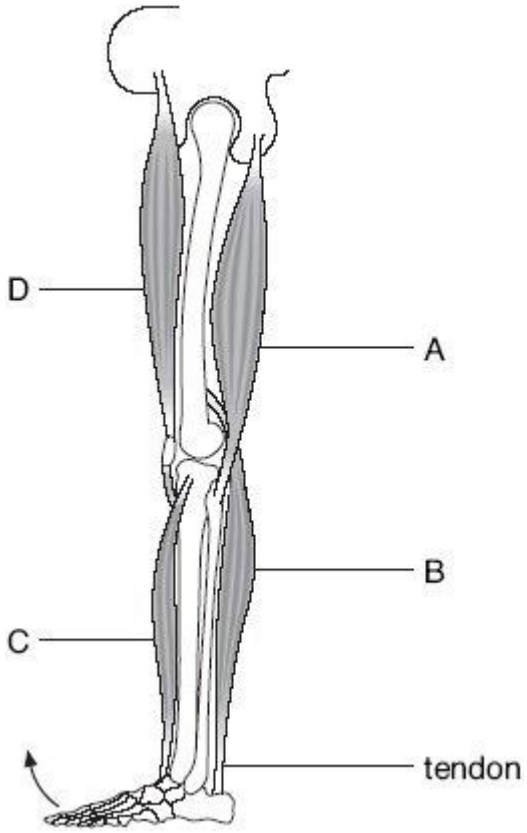
(b) The biceps and triceps are an antagonistic pair of muscles.
Explain what this means.

.....

.....

1 mark

(c) The diagram below shows muscles and bones of a human leg.



(i) Which muscle contracts to move the foot in the direction shown by the arrow?
Give the letter.

.....

1 mark

(ii) Which **two** pairs of muscles are antagonistic pairs?
Tick the **two** correct boxes.

A and B ☐

B and C ☐

C and D ☐

D and A ☐

2 marks
maximum 5 marks

Mark schemes

Q1.

(a) any **one** from

- the force of the muscle would be less **or** lost
- the (fore)arm **or** bone would not move
  *accept 'you could not bend **or** straighten your arm'*
- the biceps **or** muscle could not pull on the bone
- the biceps **or** muscle would have no effect
  *accept 'the biceps **or** muscle would not work properly'*
  'the arm would not work properly' is insufficient properly'
- the biceps **or** muscle would have less effect

1 (L6)

(b) any **one** from

- when one contracts the other relaxes
  *do not accept 'when **one** contracts the other expands*
- when the biceps contracts the forearm is raised and when the triceps contracts the forearm is lowered
  *accept 'one muscle moves the joint **or** bone **or** arm one way and the other muscle moves it the other way', 'they work together' or 'they do opposite things' is insufficient*

1 (L6)

(c) (i) C

1 (L5)

(ii) B and C ✓

1 (L5)

D and A ✓

if more than two boxes are ticked, deduct one mark for each incorrect tick minimum mark zero

1 (L5)

[5]